

Chapter 3

Assignment 3

Deadline: Feb 11, 2026

Short Answer Questions:

1. What is backend development in a web application?
2. Define MVC architecture.
3. Differentiate between frontend and backend.
4. What is a request-response cycle?
5. What is routing in web development?
6. Why is Django called a “batteries-included” framework?
7. What is the purpose of urls.py in Django?
8. What is FastAPI mainly used for?
9. What language is used in Spring Boot?
10. What is form data?
11. What is session management?
12. What is a cookie?
13. What is session expiration?
14. Name any one authentication method used in backend systems.
15. What is middleware?
16. Where does middleware sit in the request-response cycle?
17. What is logging?
18. What is error handling middleware?
19. What is CORS, CSRF and XSS?
20. Give one example of middleware usage.

Long Answer Questions:

1. Describe the role of backend in web applications. Explain how it interacts with frontend, database, and external services.
2. Describe routing, middleware, and templating concepts in backend frameworks.
3. Explain the architecture of Django framework with its major components.
4. Describe how Django handles URL routing and views.
5. Explain the overview and comparison of the following backend frameworks:
 - a. Django
 - b. Flask
 - c. FastAPI
 - d. .NET MVC

- e. Ruby on Rails
 - f. Spring Boot
 - g. Node.js
6. Describe CRUD operations with suitable examples.
 7. Explain the ORM concept. How does ORM simplify database operations?
 8. Describe sessions and cookies. How are they used to maintain user state?
 9. Compare session-based authentication and JWT-based authentication.
 10. Explain how secure login systems are implemented in backend applications.
 11. Describe middleware used for logging and error handling. Explain how middleware improves security in web applications.
 12. Explain common web security threats and how backend frameworks prevent them.